

The Restorative Influence of Virtual Reality Environment Design

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Figure 1: The screenshots featured in order from left to right include the virtual environments of the 3D moving forest, the forest image, and the abstract art. All participants that were not in the control condition were in a 3D cabin looking out into one of the above environments.

ABSTRACT

We aim to explore whether beauty in moving and still virtual environments (VEs) contributes to restorativeness. We hypothesized that the moving forest environment would be the most restorative and the abstract art would be the least restorative. The Perceived Restorativeness Scale (PRS) and the Zuckerman Inventory of Personal Reactions (ZIPERS) positive affect showed a significant increase in restorativeness in the moving forest condition than in the control condition. Additionally, PRS indicated more significance in the moving forest condition than in the abstract art condition.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Accessibility—Accessibility design and evaluation methods—; Human-centered computing—Virtual reality—;—Software and its engineering—Integrated and visual development environments—; Software and its engineering—Application specific development environments—

1 INTRODUCTION

There are two main types of stress: eustress, which is positive and motivational, and distress, which is harmful and exhausts mental resources. Distress is an increasingly prevalent issue, so effective stress management is essential for maintaining overall health [7]. Forest bathing is a particularly effective method of stress reduction, defined as immersing oneself in nature to reduce distress and restore mental resources [9]. Another restorative technique is viewing artwork, as prior research with hospitalized adults suggests it can improve individuals' moods and assist the healing process [1]. Additionally, viewing art that reflects nature has helped reduce pain in hospital patients [11]. Given the benefit of nature imagery in hospitals, more immersive nature depictions may provide benefits beyond what a wall painting offers. However, immersive nature and immersive art experiences may not be accessible or feasible for various reasons, including disabilities, work schedules, safety, and location. Recently, researchers have been interested in increasing forest bathing and art viewing accessibility via virtual reality (VR), an audio-visually immersive platform [5, 12]. This study uses VR to

compare the restorative benefit of two stress reduction techniques, aiming to understand if a restorative benefit comes from the beauty of art or nature or if more benefit is derived from the lively quality of nature. We captured nature's lively qualities in VR via moving plant life, which we compared to pictorial scenes of art and nature.

The **contributions** of our work include (1) comparing the beauty characteristics of VEs with (a) moving qualities and (b) nature objects, as well as (2) providing further evidence on both the potential restorative qualities of virtual nature environments and the effectiveness of the Masters et al. [5] implementation of the Markus and Peters Arithmetic Test (MPATest).

2 METHODOLOGY

This Institutional Review Board-approved study examined a 3D immersive moving natural environment compared to a similar 2D still-life image of a real forest and a 2D abstract art image. We **hypothesize** the following:

H_1 : The VE with moving biomass is more restorative than the other VEs and our control treatment, under the condition that the moving virtual forest most closely simulates real forest bathing [4].

H_2 : The VE with abstract art and no nature is the least restorative of the three VEs based on the prior literature comparing nature depictions to abstract art forms [10].

82 participants were recruited from Colorado State University after exclusion. Exclusion criteria included self-reported vision below 20/60 and self-reported history of heart conditions or seizures. Four participants were removed due to incomplete physiological measurements. The remaining 78 participants were aged 18 to 35, with a mean of 21. 31 identified as female, and the rest identified as male. 47.4% of the pool reported wearing corrective lenses.

The experiment was a between-subjects study using Latin squares random assignment to one of four conditions: VR moving nature, VR nature image, VR abstract art image, or control. After consenting and completing demographic forms, participants underwent initial physiological and psychological measurements, including these measures in order: Heart Rate (HR), Blood Pressure (BP), the Positive and Negative Affect Schedule (PANAS), ZIPERS, and PRS. Afterward, participants completed the MPATest, a mathematical test to induce stress. The measures were retaken to verify the efficacy of the stressor test. Participants then spent ten minutes in their assigned environment, followed by completing the final set of measures. They then completed a preference questionnaire, ranking nature images similar to the VEs, and an open-exit questionnaire asking about cybersickness and what they did and did not like. The participants were then debriefed and dismissed.

All VEs were designed using Unreal Engine 4 [2] with identical room layouts and forest audio [8], the only difference being the

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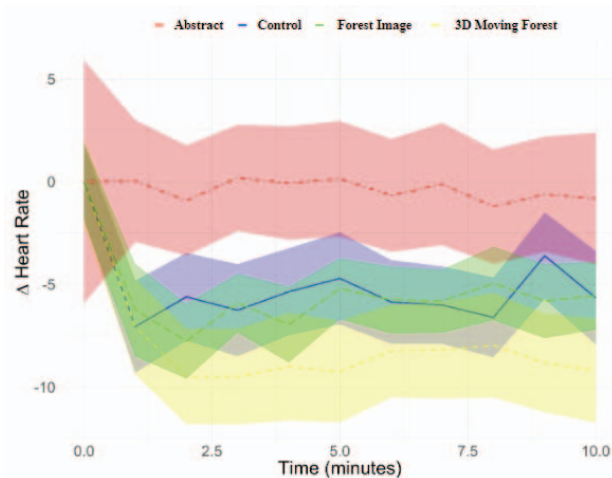


Figure 2: The average participant's HR every minute as they experienced each experimental treatment for ten minutes.

window view, which showed either an abstract art image, nature image, or 3D nature simulation. Each participant sat in a chair looking toward the window to minimize cybersickness [3].

The piece chosen for the abstract art VE was Piet Mondrian's Composition II in Red, Blue, and Yellow, 1930 [6] because it captured artistic beauty while lacking colors reflective of a forest. In the forest image VE, a photograph was selected to capture the beauty of real nature in a stationary image, including rolling mountains, a top view of trees, and rocks covered in snow. The third VE option was the 3D moving forest designed to look similar to the forest image yet have the moving qualities of being immersed in nature using a collection of virtual trees and mountains with movement and depth. In the control condition, participants closed their eyes for ten minutes to model a way of escaping stressful visual stimuli and relaxing without VR treatment.

3 RESULTS AND DISCUSSION

T-tests were used to determine the efficacy of the stress test on all participants across conditions (pre-stress minus post-stress comparison). The change in participant stress levels (post-environment minus post-stress) was measured between conditions after completing the experimental treatments using Analysis of Variance (ANOVA) tests for HR, positive/negative ZIPERS, positive/negative PANAS, and PRS. A Multivariate ANOVA (MANOVA) test was used for BP measurements (post-stress post-environment comparison). The MANOVA test was conducted on BP because diastolic and systolic BP depend on each other for each participant between independent conditions.

Except for diastolic BP and PRS, all dependent variables supported the MPATest's effectiveness, consistent with Masters et al.'s results [5]. There are many possible reasons that diastolic BP did not provide significance, such as participant characteristics. The questions on PRS do not apply to the stressor experience and were administered to maintain consistency in measures.

Although the physiological measurements did not demonstrate significance for the restorativeness of any conditions, important HR patterns could be observed. These patterns are shown in Figure 2, demonstrating that our 3D moving forest consistently provided the lowest HR measurements. Regarding psychological measurements, PRS found significance between the 3D moving forest VE and the abstract art and control conditions. Additionally, the ZIPERS positive effect showed significance between the 3D moving forest

VE and the control condition. Thus, PRS and ZIPERS positively support h_1 that the 3D moving VE is more restorative than the abstract art and control conditions. The preference questionnaire responses potentially explain why the 2D still forest VE did not show a significant difference, as the images of the real environments were consistently rated higher than the screenshot of the 3D moving forest VE.

H_2 is partially supported because the abstract art condition significantly differed from the 3D moving forest VE but not from any other condition. Our results align with the preference questionnaire results, as the abstract art image was the lowest rated. Further, multiple aspects of our thematic analysis for the open-exit questionnaire helped support h_1 and h_2 . Multiple comments suggested the participants preferred the 3D moving forest and forest image conditions over the abstract art condition.

4 CONCLUSION

Limitations of this study included the limited range of demographic differences that may be uncontrolled since we utilized Latin square assignment, lack of active movement and engagement, and potential inaccuracies in the BP readings due to factors like the amount of time spent sitting. Future study topics include user movement and interaction, the effects of cybersickness, and additional open-response surveys to understand user perception of the environment further. Results from this study indicate that tactfully designed VEs have the potential for reducing stress, but further investigation is needed into how to design for optimal stress reduction.

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